IGCSE 0607 Practice Questions — E9.3

Probability of combined events

Total Questions: 9 — **Total Marks:** 60 Marks

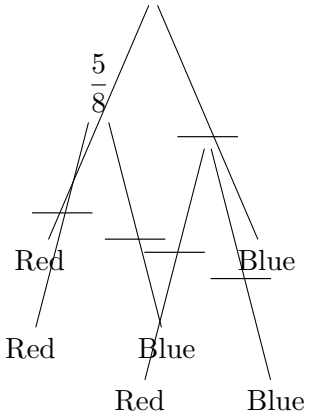
Table of Contents & Quick Navigation

- **E9.3** Probability of combined events
  → **Paper 2 (Non-Calculator)** *(5 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

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1 A bag contains 5 red counters and 3 blue counters.
A counter is taken at random from the bag, its colour is noted and it is replaced.
A second counter is then taken at random.

(a) Complete the tree diagram.



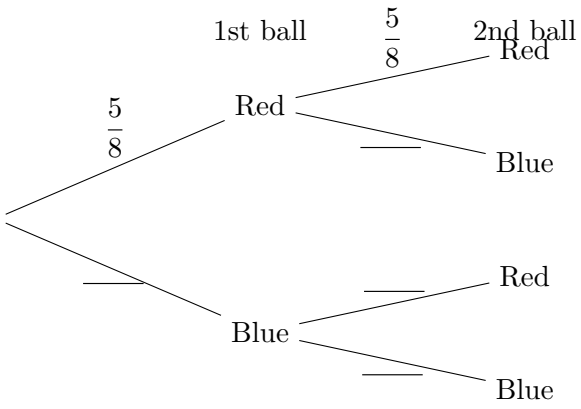
[2]

(b) Calculate the probability that both counters are red.

..... **[2]**

2 A bag contains 8 balls.
5 of the balls are Red and 3 of the balls are Blue.
A ball is taken at random from the bag, its colour is noted, and it is replaced.
A second ball is then taken at random from the bag.

(a) Complete the tree diagram.



[2]

(b) Calculate the probability that both balls are Red.

..... **[1]**

(c) Calculate the probability of picking one Red ball and one Blue ball in any order.

..... **[2]**

3 A box contains n counters.
3 of the counters are Red and all the remaining counters are Blue.
Two counters are taken at random from the box, without replacement.
The probability that both counters are Red is $\frac{1}{15}$.

(a) Show that $n^2 - n - 90 = 0$.

[3]

(b) Find the value of n.

$n =$ ..... **[2]**

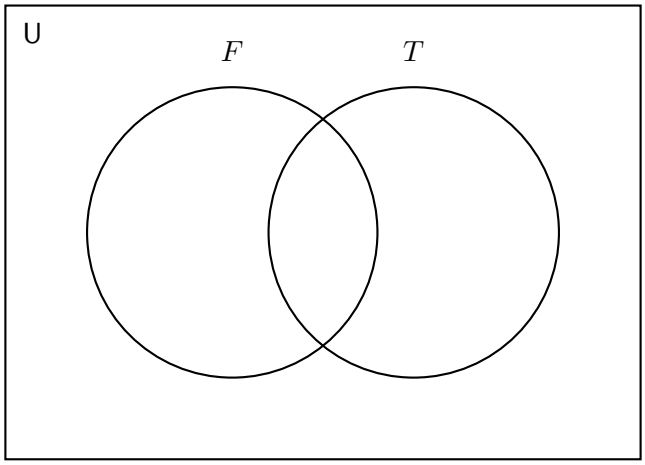
(c) Calculate the probability that exactly one of the two counters chosen is Red.

..... **[3]**

4 There are 30 students in a class.
U = {students in the class}
F = {students who play Football}
T = {students who play Tennis

18 students play Football. 14 students play Tennis. 4 students do not play Football and do not play Tennis.

(a) Complete the Venn diagram to show the number of students in each region.



[2]

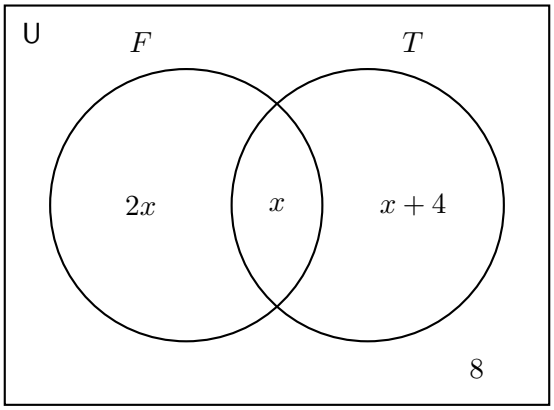
(b) A student is chosen at random from the class.
Find $P(F \cap T)$.

$P(F \cap T) =$ ..... **[1]**

(c) Two students are chosen at random from the class.
Calculate the probability that both students play Football but not Tennis.

..... **[3]**

5 A group of 40 students are asked if they play Football (F) or Tennis (T).
The Venn diagram shows the number of students in each region in terms of x.



(a) Find the value of x.

$x =$ ..... **[2]**

(b) Find $P(F \cup T)$.

..... **[1]**

(c) Two students are chosen at random from the group of 40.
Calculate the probability that both students play Football only.

..... **[3]**

6 A bag contains 4 red discs and some blue discs.
Let the total number of discs in the bag be x.
Two discs are taken at random from the bag, without replacement.
The probability that both discs are red is $\frac{1}{11}$.

(a) Show that $x^2 - x - 132 = 0$.

[3]

(b) Find the number of blue discs in the bag.

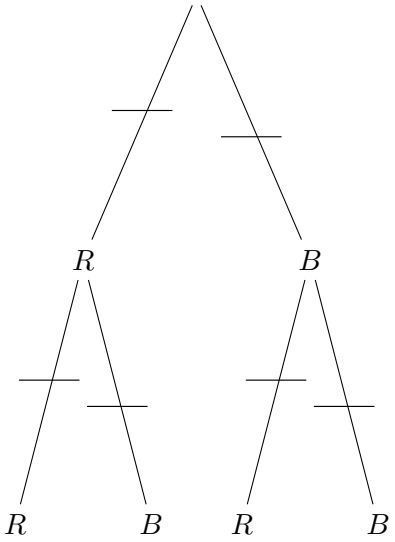
..... **[3]**

(c) Calculate the probability that one disc is red and one disc is blue.

..... **[3]**

7 A bag contains 7 red counters and 5 blue counters.
A counter is taken at random, its colour is noted, and it is NOT replaced in the bag.
A second counter is then taken at random and its colour is noted.

(a) Complete the tree diagram by writing the missing probabilities on the branches.



[3]

(b) Calculate the probability that both counters are red.

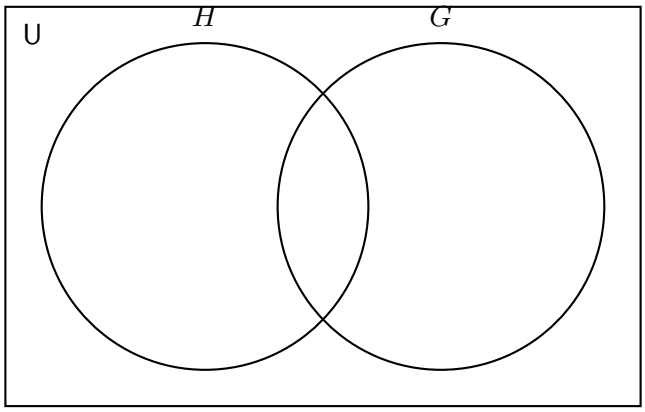
..... **[2]**

(c) Calculate the probability that one counter is red and one counter is blue.

..... **[2]**

8 In a group of 50 students, 32 study History (H), 21 study Geography (G), and 6 study neither subject.

(a) Complete the Venn diagram to show the number of students in each region.



[3]

(b) A student is chosen at random from the group of 50.

Find:

(i) $P(H \cap G)$

$P(H \cap G) =$ ..... **[1]**

(ii) $P(H \cup G')$

$P(H \cup G') =$ ..... **[1]**

(c) Two students who study History are chosen at random.
Calculate the probability that both students also study Geography.

..... **[2]**

9 A box contains x red balls and 4 blue balls.
Two balls are taken from the box at random, without replacement.
The probability that both balls are red is $\frac{1}{3}$.

(a) Show that $x^2 - 5x - 6 = 0$.

[3]

(b) Work out the number of red balls in the box.

..... **[2]**

(c) Calculate the probability that the two balls taken have different colours.

..... **[3]**